$A \in M_n(\mathbb{R})$

$\lambda = \rho \cdot e^{i\theta}$ $\quad$ due autovalori complessi

$\bar{\lambda} = \rho e^{-i\theta}$

$f: \mathbb{R}^n \to \mathbb{R}^n \qquad M_n(f) = M_n(g) = A$

$g: \mathbb{C}^n \to \mathbb{C}^n$

$K = e_1, \dots, e_n$ base canonica di $\mathbb{R}^n$

$K' = e_1, \dots, e_n$ " " di $\mathbb{C}^n$

$\mathbb{R}^n = \mathbb{C}^n$

sia $v \in \mathbb{C}^n$ autovettore per g relativo a λ

$g(v) = \lambda v$

$Av = \lambda v$ $\quad$ quindi $\bar{v} \in \mathbb{C}^n$ è autovettore per g

$A\bar{v} = \bar{\lambda}\bar{v}$ $\quad$ relativo a $\bar{\lambda}$

$W = \langle v, \bar{v} \rangle_{\mathbb{C}} \subseteq \mathbb{C}^n$

$w_1 = \frac{v+\bar{v}}{2} \qquad w_2 = \frac{v+\bar{v}}{2i}$

$w_1, w_2 \in \mathbb{R}^n \subseteq \mathbb{C}^n$

w_1, w_2 formano una base di spazio W

$$f(w_1) = \frac{g(v)+g(\bar{v})}{2} = \frac{\lambda v + \bar{\lambda}\bar{v}}{2} = \frac{\left(\rho(\cos\theta + i\sin\theta)\, v + \rho(\cos\theta - i\sin\theta)\bar{v}\right)}{2} =$$

$$= \rho\cos\theta\, \frac{v+\bar{v}}{2} + i\rho\sin\theta\, \frac{v-\bar{v}}{2} = \rho\cos\theta\, w_1 - \rho\sin\theta\, w_2$$

$$f(w_2) = \frac{g(v)-g(\bar{v})}{2i} = \frac{\lambda v - \bar{\lambda}\bar{v}}{2i} = \frac{\rho(\cos\theta - i\sin\theta)\cdot v - \rho(\cos\theta - i\sin\theta)\bar{v}}{2i} =$$

$$= \frac{\rho\cos\theta}{2i}(v-\bar{v}) + \frac{\rho\sin\theta}{2}(v+\bar{v}) = \rho\cos\theta\, w_2 + \rho\sin\theta\, w_1$$

$$M_{\mathcal{C}}(f_{|w}) = \begin{pmatrix} \rho\cos\theta & \rho\sin\theta \\ -\rho\sin\theta & \rho\cos\theta \end{pmatrix} = \rho \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

(calcolata su $\mathcal{C}$)